

# MedTalk: Explanation-Driven Adaptive Questioning for Responsible Interpretable AI Across Regulated Sectors

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**Abstract**—With the growing use of machine learning (ML) models in critical applications, the issue of the “interpretability gap” continues to pose a considerable obstacle to both trust and adherence to regulatory standards. Existing conversational explainable AI (XAI) systems, such as TalkToModel (TTM), exhibit vulnerabilities inherent in Large Language Models (LLMs), which include hallucinations and suboptimal probability calibration. We introduce MedTalk, a four-layered deterministic framework that replaces stochastic reasoning with transparent Random Forest classifiers and a novel Explanation-Driven Adaptive Questioning (EDAQ) algorithm. EDAQ proactively guides dialogues by quantifying feature uncertainty and significance. In experiments using the UCI Pima Indian Diabetes dataset, MedTalk achieved 89.4% accuracy and a Brier score of 0.142, significantly outperforming TTM’s calibration (0.189). A clinical user study ( $N = 32$ ) demonstrated a 21-point improvement in user comprehension (94.2% vs. 73.1%). Our results confirm MedTalk as a robust architecture for responsible AI in regulated sectors.

**Index Terms**—Explainable AI, Black Box Models, SHAP, LIME, Pima Indian Diabetes, TalkToModel, EDAQ.

## I. INTRODUCTION

Highly accurate predictions have been made by machine learning models in domains like healthcare, finance, and law. The complexity of these models creates difficulty in interpretation [1], [3]. This lack of interpretability has created an *interpretability gap* which creates a barrier for stakeholders to establish trust, fairness and regulatory compliance [2], [5]. In high-stake applications, opaque predictions could compromise clinical safety or create systemic bias, therefore creating a need for regulatory requirements for auditable and explainable artificial intelligence (AI) systems [2]–[5].

Post-hoc explanation methods, such as SHAP [7] and LIME [8] are used to create feature level explanations. A post-hoc explanation method called Talk To Model (TTM), allows users

to interactively create explanations about model outputs via natural language dialog [14]. There is however a limitation to TTM, it uses generative models that can hallucinate [17] produce miscalibrated probability estimates [18] and remains purely reactive [19], [20].

To improve upon the current limitations of TTM, we propose **MedTalk**, a deterministic conversational XAI framework that uses an *Explanation Driven Adaptive Questioning* (EDAQ) algorithm. EDAQ uses SHAP values and ensemble variance to identify features that are both influential and uncertain, and then generates targeted questions to refine the predictions made by MedTalk. We tested MedTalk’s effectiveness using the UCI Pima Indian Diabetes dataset [21]. Our results show that MedTalk was able to reach 89.4

## II. THE PREDICAMENT OF PREDICTIVE OPACITY

The swift advancement of machine learning has encountered a contradictory point where predictive accuracy often comes at the cost of decision clarity. Contemporary computational approaches, especially deep neural networks and gradient-boosted ensembles, are described as “black boxes” since their internal transformations consist of complex parameter interactions that are not easily understood by humans [1], [2]. While these models achieve state-of-the-art results, their inherent opacity creates an interpretability gap that is particularly problematic in regulated sectors [3], [5].

In the field of healthcare, depending on opaque models can pose serious risks to life. A model may detect a “pneumonia risk” pattern based on clinically insignificant artifacts instead of physiological indicators [3], [4]. In the absence of methods to explore the reasoning behind these models, clinicians must choose between placing blind trust in them or dismissing potentially helpful technology [3]. Likewise, in the finance

sector, non-transparent algorithms can unintentionally embed systemic biases, which can breach fair-lending laws [1], [5]. To achieve transparency, it is essential to create interfaces that connect complex mathematical concepts with the cognitive requirements of humans [1], [2].

### III. FOUNDATIONS OF POST-HOC INTERPRETABILITY

To address the black box dilemma, several explainability algorithms have been developed to provide insights into model behavior after training [5]–[7].

#### A. Local Interpretable Model-Agnostic Explanations (LIME)

LIME operates on the principle of local surrogate modeling. It assumes that while the global decision boundary may be intricate, it can be approximated by a linear model in the immediate vicinity of a specific data point [6], [8]. By generating perturbed samples and querying the black box, LIME builds a simpler, interpretable model to explain individual outcomes [8].

#### B. SHapley Additive exPlanations (SHAP)

Based on cooperative game theory, SHAP considers features as “players” and the prediction as a “payout” [7], [9]. The Shapley value for a feature represents its average marginal contribution across all possible feature coalitions, adhering to essential axiomatic properties such as symmetry and additivity [7], [10], [11].

#### C. Gradient-Based XAI Methodologies

For differentiable models, gradient based methods such as Integrated Gradients (IG), provide direct insight by utilizing internal gradients [11], [12] that are used to integrate the gradients along a path from a neutral baseline to the input in question; the sum of all the attributions will equal the total difference between the actual model output and the baseline [10], [11], [13].

### IV. PROPOSED METHOD

#### A. Architecture Overview

MedTalk is structured around **four layers that operate independently of any particular domain**:

- 1) **Prediction Layer:** A Random Forest classifier comprising 500 trees with a maximum depth of 18, optimized for tabular inputs.
- 2) **Attribution Layer:** Feature attribution is produced using SHAP (TreeSHAP) in combination with LIME, with an automatic mechanism to choose the explanation method yielding the highest fidelity.
- 3) **Orchestration Layer:** The Explanation-Driven Adaptive Questioning (EDAQ) module—detailed in Algorithm 2—coordinates follow-up questioning based on model uncertainty and key feature contributions.
- 4) **Interface Layer:** A conversational interface that incorporates sector-specific rules and operational constraints.

**Design Invariant:** Model outputs always stem from *transparent and deterministic* machine-learning components. Adaptation to new sectors is handled through configurable constraint templates rather than through learned or stochastic policies.

#### B. Cross-Domain Instantiation

##### C. Implementation Across Domains

MedTalk can be customized for different areas with only a few essential elements:

- (i) a set of question formats,
- (ii) directives that specify the restrictions particular to each industry, and
- (iii) a retrieval system for obtaining context that pertains to each specific domain.

Importantly, these elements allow for deployment without having to retrain the core model.

- **Healthcare:** The templates relate to standard clinical questions, and the protocols guarantee that inputs stay within medically acceptable boundaries.
- **Finance:** The templates align with standard underwriting questions, whereas the guidelines aim to avoid biased practices such as redlining.

### V. VULNERABILITIES IN STRUCTURAL ARCHITECTURE OF TALKToMODEL

TalkToModel (TTM) by Slack et al. (2023) has been recognized as a pioneering work in the area of conversational XAI [14], [15]. Nevertheless, various structural vulnerabilities have been identified in the architecture of TTM, which can limit its ability to be used in high-risk domains [14], [16]. The use of large language models (LLMs) for both interpretation and generation of narratives creates a significant risk of “hallucinations” i.e., the system generates explanations that are not based upon the actual logic of the underlying machine learning model [3], [17], [18]. Further, TTM does not provide any proactive guidance and is instead a reactive entity that only responds to the questions asked of it, potentially failing to identify important uncertainty regions [19], [20]. Additionally, the probabilistic nature of TTM makes it incompatible with the deterministic auditability requirements mandated by regulatory agencies [1], [2].

### VI. THE PROPOSED MEDTALK ARCHITECTURE

MedTalk substitutes stochastic LLM reasoning with a four-layer deterministic framework designed for auditability and regulatory compliance:

- **Prediction Layer:** A transparent Random Forest classifier comprising 500 estimators with a maximum depth of 18, optimized for tabular clinical inputs [3], [4].
- **Attribution Layer:** A hybrid approach that computes both SHAP and LIME fidelity scores for every instance, automatically selecting the explanation method with the highest local faithfulness [7], [8].

- **Orchestration Layer:** Contains the Explanation-Driven Adaptive Questioning (EDAQ) algorithm to steer dialogues based on model uncertainty and feature relevance [19], [20].
- **Interface Layer:** Employs domain-specific constraint templates for adapting responses to sector requirements while maintaining regulatory compliance [2], [5].

**Design Invariant:** Model outputs always stem from *transparent and deterministic* machine-learning components. Adaptation to new sectors is handled through configurable constraint templates rather than through learned or stochastic policies.

## VII. THE EDAQ ALGORITHM

The primary innovation is the Explanation-Driven Adaptive Questioning (EDAQ) module. EDAQ calculates an uncertainty score  $U(f)$  for each feature:

$$U(f) = \text{MissRate}(f) + 0.5 \times \text{Variance}(\text{ModelEnsemble}(x)) \quad (1)$$

The algorithm prioritizes questions by focusing on features that are both influential (high SHAP value) and uncertain ( $U(f)$ ) [7], [19]. It then filters results to focus on actionable factors, passing every query through fairness guardrails to prevent discriminatory inquiries [2], [5].

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### Algorithm 1 Explanation-Driven Risk Assessment

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**Require:** Feature vector  $x$ , trained model  $f$ , feature names  $F$ , domain type  $D$

**Ensure:** Risk prediction  $\hat{y}$ , explanation dict  $E$ , follow-up questions  $Q$

```

1:  $\hat{y} \leftarrow f.\text{predict\_proba}(x)[1]$ 
2:  $\text{risk\_cat} \leftarrow \text{categorize\_risk}(\hat{y}, D)$   $\triangleright$  Domain thresholds
3:  $x_{\text{trans}} \leftarrow \text{preprocess}(x, D)$ 
4:  $\triangleright$  Compute SHAP + LIME
5:  $\text{explainer} \leftarrow \text{TreeExplainer}(f)$ 
6:  $\phi \leftarrow \text{explainer.shap\_values}(x_{\text{trans}})$ 
7:  $E_{\text{shap}} \leftarrow \{F[i] : \phi[i] \text{ for } i \text{ in } 1..d\}$ 
8:  $E_{\text{lime}} \leftarrow \text{LIME}(f, x, \text{num\_samples} = 1000)$ 
9:  $\triangleright$  Select most faithful explanation
10:  $\text{faith} \leftarrow \text{compute\_fidelity}(E_{\text{shap}}, E_{\text{lime}}, f, x)$ 
11: if  $\text{faith}[\text{SHAP}] > \text{faith}[\text{LIME}]$  then
12:    $E \leftarrow E_{\text{shap}}$ 
13: else
14:    $E \leftarrow E_{\text{lime}}$ 
15: end if
16:  $\triangleright$  Extract top features and domain context
17:  $\text{top\_features} \leftarrow E.\text{top\_k}(k = 5)$ 
18:  $\text{domain\_context} \leftarrow \text{lookup\_context}(\text{top\_features}, D)$ 
19:  $\triangleright$  Generate adaptive follow-up questions
20:  $Q \leftarrow \text{EDAQ}(\text{top\_features}, \text{domain\_context}, x, D)$ 
21: return  $\hat{y}, \text{risk\_cat}, E, \text{domain\_context}, Q$ 

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### A. Algorithm 1: Explanation-Driven Algorithm For Risk Assessment

Algorithm 1 is the main entry for all MedTalk Inference requests. It takes a feature vector  $x$  and a pre-trained Random Forest Model  $f$  as inputs. The algorithm begins by calculating the calibration of the prediction probability  $\hat{y}$  with the models `predict_proba()` method. Then, it maps the predicted probability to a risk level (Low, Medium, High) based on the application defined threshold levels, to create risk communication that is understandable from a clinical perspective.

Feature attributions are also calculated to create an explanation of the predictions made by the model. The two methods used for this purpose are TreeSHAP and LIME. TreeSHAP provides exact SHAPley values explanations through the use of cooperative game theory, while LIME creates an approximate linear surrogate model of the model's behavior at a specific point in space, but does so in a way that is local to that point. The algorithm will evaluate the fidelity of both types of explanations at the point in space of interest and select the type of explanation that best represents the model's behavior at that point.

The attributes with the greatest influence on the model's prediction are identified and further enriched with additional domain-specific context information. The resulting set of features is passed to the Explanation-Driven Adaptive Questioning (EDAQ) algorithm, which uses the features to generate adaptive follow-up question(s), to reduce the uncertainty associated with the prediction. The EDAQ algorithm produces the original prediction, risk category, an explanation dictionary, and the generated question(s) and passes them back to the User Interface.

### B. Algorithm 2: Interactive Refining Process With Fairness Constraints (guardrails)

Algorithm 2 provides a framework for the back-and-forth exchange between the system and the end-user during the multiple turns of the interaction loop. Following the first prediction and the generation of an explanation, EDAQ generates the clarification questions. The user's response is used as input to update the feature vector; the prediction is then recalculated.

The iterative refining process will be repeated for a few iterations or until there is no further need for refining. All modifications to the feature vector are documented in an immutable audit log containing the name of the feature, the previous value of the feature, the new value of the feature and the iteration at which the changes were made. This design allows for a complete audit trail of the process leading to the conclusion, which is required by law for all audits.

In order to provide fairness constraints on each feature update the domain specific constraint function is applied. If a proposed update to the feature violates either fairness or regulation rules — such as adding the protected demographics to a lending decision the system immediately terminates the reasoning path and logs the violation. These safeguard mechanisms ensure that discrimination does not influence the reason and/or final decision.

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**Algorithm 2** Interactive Refinement with Fairness Guardrails

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**Require:**  $\hat{y}_0, E_0, Q$ , user responses  $R$ , domain  $D$ **Ensure:** Refined prediction  $\hat{y}_{\text{final}}$ , audit log

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1: audit_log  $\leftarrow \emptyset$ ,  $y_{\text{curr}} \leftarrow \hat{y}_0$ ,  $x_{\text{curr}} \leftarrow x$ 
2: violations  $\leftarrow 0$ , fairness_budget  $\leftarrow 0.05$ 
3: for iter in 1..3 do
4:   for each  $q$  in  $Q$  do
5:     if  $q.\text{feature} \in R$  then
6:        $x_{\text{curr}}[q.\text{feature}] \leftarrow R[q.\text{feature}]$ 
7:       audit_log.append(feat, old, new, iter)
8:     end if
9:   end for
10:    $\triangleright$  Check for fairness violations
11:   for each (feat, new_val) in  $R$  do
12:     if violates_fair_lending(feat, new_val,  $D$ ) then
13:       violations += 1
14:       STOP  $\triangleright$  Prevent discriminatory trajectory
15:     end if
16:   end for
17:    $y_{\text{new}} \leftarrow \text{model.predict_proba}(x_{\text{curr}})[1]$ 
18:   if  $|y_{\text{new}} - y_{\text{curr}}| < 0.01$  then break
19:   end if
20:    $E_{\text{new}}, Q_{\text{new}} \leftarrow \text{Alg1}(x_{\text{curr}}, \text{model}, D)$ 
21:    $y_{\text{curr}} \leftarrow y_{\text{new}}$ 
22: end for
23: log_prediction( $y_{\text{final}}, x_{\text{final}}, E_{\text{final}}, \text{audit\_log}, D$ )
24: return  $y_{\text{curr}}, E_{\text{final}}, \text{audit\_log}$ 
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*C. Algorithm 3: Explanation-Driven Adaptive Questioning (EDAQ)*

Algorithm 3 is the primary mechanism to convert feature attributions to action-oriented follow-up questions. By combining model importance scores with uncertainty estimates, EDAQ determines which features need clarification from the user.

Firstly, EDAQ calculates an uncertainty score for each feature based on missing data rates and model prediction agreement among the ensemble models. Those features having high uncertainty can indicate that gathering further information could substantially improve predictive reliability.

Secondly, EDAQ employs several constraint filters to eliminate those features that do not qualify as candidates for clarification. These include features of relatively low impact, legally protected attributes, and those features that cannot be modified by the user or observed by them. The remaining candidate features are then ranked according to their joint score of attribution magnitude and uncertainty.

Thirdly, for those candidate features that have the largest joint scores, EDAQ creates domain-specific questions based upon previously defined semantic question templates. Each question includes context to explain why the feature is important to the currently predicted value. Fourthly, EDAQ performs a fairness and regulatory compliance check to ensure that none of the generated questions indirectly violate fairness or

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**Algorithm 3** Explanation-Driven Adaptive Questioning (EDAQ)

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**Require:** Top- $K$  features, top\_k, domain context  $C$ , features  $x$ , domain  $D \in \{\text{healthcare, finance, judiciary, insurance}\}$ ,  $\tau_u = 0.3$ ,  $\tau_i = 0.05$ ,  $Q_{\text{max}} = 5$ **Ensure:** Prioritized list of follow-up questions  $Q$ 

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1:  $Q_{\text{cand}} \leftarrow \emptyset$ , feat_uncertainty  $\leftarrow \{\}$ 
2: constraints  $\leftarrow \text{load\_constraints}(D)$ 
3: Step 1: Compute uncertainty for each feature
4: for each feature  $f$  in  $x$  do
5:   miss_rate  $\leftarrow \text{count\_missing}(f) / \text{len}(x)$ 
6:   var  $\leftarrow \text{variance}(\text{model\_ensemble.predict}(x))$ 
7:   feat_uncertainty[ $f$ ]  $\leftarrow \text{miss\_rate} + 0.5 \times \text{var}$ 
8: end for
9: Step 2: Filter by constraints and modifiability
10: modifiable_high_impact  $\leftarrow \emptyset$ 
11: for each (feat, shap_val) in top_k do
12:   if  $|\text{shap\_val}| < \tau_i$  then continue
13:   end if
14:   if is_protected(feat,  $D$ ) then continue
15:   end if
16:   if  $\neg \text{is\_modifiable}(\text{feat}, D)$  then continue
17:   end if
18:   if feat_uncertainty[feat]  $> \tau_u$  then
19:     score  $\leftarrow |\text{shap\_val}| \times \text{feat\_uncertainty}[\text{feat}]$ 
20:     modifiable_high_impact.append((feat, score))
21:   end if
22: end for
23: Step 3: Cross-domain semantic mapping
24:  $q_{\text{map}} \leftarrow \{\text{"age"} : \{\text{health} : \text{"How old are you?"}, \text{fin} : \text{"What is the applicant's age?"}\}, \dots\}$ 
25: Step 4: Multi-objective prioritization
26: sorted  $\leftarrow \text{sort}(\text{modifiable\_high\_impact}, \text{desc})$ 
27: Step 5: Generate domain-specific questions
28: for each (feat, prio) in sorted[1 :  $Q_{\text{max}}$ ] do
29:   template  $\leftarrow q_{\text{map}}[\text{feat}][D]$ 
30:   if  $D = \text{"healthcare"}$  then
31:     expl  $\leftarrow \text{f"Affects risk by } |\text{shap}| \text{"}$ 
32:     ctx  $\leftarrow \text{med\_interp}(\text{feat}, x[\text{feat}])$ 
33:   else if  $D = \text{"finance"}$  then
34:     expl  $\leftarrow \text{f"Influences approval by } |\text{shap}| \text{"}$ 
35:     ctx  $\leftarrow \text{check\_fair\_lending}(\text{feat}, D)$ 
36:   end if
37:    $Q_{\text{cand}}.\text{append}(\{q : \text{template}, p : \text{prio}, \text{ctx} : \text{ctx} \dots\})$ 
38: end for
39: Step 6: Compliance check
40:  $Q_{\text{compliant}} \leftarrow \emptyset$ 
41: for each  $q$  in  $Q_{\text{cand}}$  do
42:   if passes_fairness_check( $q, D$ ) then
43:      $Q_{\text{compliant}}.\text{append}(q)$ 
44:   end if
45: end for
46: return  $Q_{\text{compliant}}$ 
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regulatory requirements. Lastly, the resulting ordered list of questions is sent back to the interaction layer to guide the refinement process.

## VIII. EXPERIMENTAL EVALUATION

### A. Dataset: UCI Pima Indian Diabetes

To determine how well our model works in relation to this problem space, we have selected a data set from the University of California at Irvine (UCI) Machine Learning Data Repository [22] that is called the "Pima Indian Diabetes" dataset [21]. The dataset is a collection of 768 examples representing female Pima Indians; each example was created using a number of diagnostic measures such as plasma glucose concentrations, BMI, insulin levels, and the diabetes pedigree function [23].

### B. Comparative Performance Analysis

MedTalk was tested against both TTM and Baseline in order to evaluate MedTalk's performance, which is listed in Table I. MedTalk performed significantly better than TTM and Baseline models in terms of Brier scores, achieving a brier score of 0.142 while TTM scored 0.189 and logistic regression scored 0.201 [8], [14]

TABLE I  
PERFORMANCE COMPARISON ACROSS SYSTEMS

| Metric            | MedTalk      | TTM   | Log. Reg. |
|-------------------|--------------|-------|-----------|
| Accuracy          | <b>0.894</b> | 0.812 | 0.798     |
| Brier Score       | <b>0.142</b> | 0.189 | 0.201     |
| Clin. Comp. %     | <b>94.2%</b> | 73.1% | N/A       |
| Trust (Likert /7) | <b>6.2</b>   | 4.1   | N/A       |
| Task Time (s)     | <b>63.2</b>  | 134.7 | N/A       |

The results show that MedTalk is able to predict with an accuracy of 89.4 %, which is higher than both TTM (81.2%) and Logistic Regression (79.8%); however, it is important to note that MedTalk has the smallest Brier Score (0.142) and thus has significantly better predictive probability calibration than both TTM (Brier Score of 0.189) and Logistic Regression (Brier Score of 0.201). The improvements in MedTalk's prediction results can be attributed to MedTalk's use of a Deterministic Random Forest for prediction as opposed to LLM-generated Confidence Estimates that are uncalibrated statistically [2], [14].

### C. Clinical User Study

User studies have shown a significant improvement in terms of usability and explainability in clinical settings. In comparison to TTM, MedTalk demonstrated a much better usability score with a 94.2% vs 73.1% success rate in completing tasks [20], which corresponds to a 21 point increase. The time spent on completing tasks was reduced from 63.2 seconds to 134.7 seconds as a result of the MedTalk system, demonstrating that EDAQ's method of providing feedback to users through questions can reduce the effort needed to understand model

predictions [19]. The trust scores provided by the users were significantly higher for MedTalk than for TTM. Users rated the MedTalk explanation at an average score of 6.2 out of 7, while they rated the TTM explanation at 4.1 out of 7 [17]. This is expected because LLM-based explanations may be susceptible to hallucinations [17].

### D. Ablation Study

Ablation studies confirm the contribution of each MedTalk component. Removing EDAQ drops our comprehension from 94.2% to 81.3%, underscoring that proactive questioning was the main contributor to interpretability gains. Removing the hybrid explanation mechanism has a small detrimental effect on calibration, whilst disabling fairness guardrails resulted in 7.3% of questions referring to demographic proxies in financial testing tasks [5].

## IX. RELATED WORK

### A. Dialogue Systems for AI Explainability

*TalkToModel* [14] represents the first approach to query machine learning models using natural language dialog. The authors integrate a T5-based semantic parser with a BART-based response generator. According to Slack et al. (2023), TalkToModel achieves an average parsing accuracy of about 76%. In addition, according to Slack et al. (2023), 82% of the healthcare practitioners who participated in the survey preferred to use TalkToModel's dialog interface to access explanations of their model's output compared to the traditional dashboard-style interfaces that have been used up until now [3], [14].

**Limitations** Despite the fact that TTM has had such a major influence on the way we think about model interpretation, TTM is still missing one important component, which is probability calibration analysis; this is very important when you are working in a risk-sensitive domain. Because of the fact that TTM uses generative Large Language Models (LLMs), it can behave in ways that are unpredictable. TTM is mainly limited to supervised tabular tasks. Finally, there are no features in TTM that would allow users to be assured that they are meeting all the relevant regulatory requirements [16]–[18].

**Other Dialogue-Adjacent Tools:** The Language Interpretability Tool (LIT) [13] provides visualization and interpretability capabilities for NLP models, while the What-If Tool [12] allows for the execution of counterfactual analyses. Neither tool offers the ability to support interactive multi-turn dialogue or deployment frameworks for regulated industries [19], [20].

### B. Explainable AI in High-Stakes Domains

**Healthcare:** Caruana et al. (2015) [3] have been at the forefront of developing interpretable ML technology that will assist in making medical decisions based upon a patient's history and symptoms; particularly with respect to pneumonia risk prediction. Rajkomar et al. (2018) [4] developed extensions of interpretability techniques for use with deep learning models that were trained using Electronic Health

Records (EHRs). Today’s literature emphasizes additional factors regarding the development of new ML systems including calibration reliability [5], equitable forecasting [2] and fairness across demographic groups [5].

**Finance:** Khandani et al. [13] conducted early work illustrating the utility of ensemble methods in evaluating credit risk, and today’s U.S. Securities and Exchange Commission (SEC) regulations require that decision-making associated with algorithmic trading be transparent and auditable.

Barocas and Self (2016) [6] analyzed the systemic equity issues embedded within ML-based lending frameworks.

**Judiciary:** Dressel and Farid (2018) [14] identified significant bias in widely used risk assessment tools such as COMPAS and argued that there should be alternative tools that are transparent and verifiable. Corbett-Davies et al. (2016) [10] also illustrated how fairness and accuracy objectives often conflict when it comes to criminal justice applications.

**Recognized Gap:** The majority of prior research focuses on domain-specific solutions, resulting in fragmented interpretability efforts. No existing overarching framework provides transparent, conversation-driven explanations that dynamically adapt to the safety constraints and regulatory requirements of multiple sectors simultaneously [5], [19], [20].

## X. CONCLUSIONS

MedTalk demonstrates that a deterministic orchestration of transparent sub-components offers a more reliable and auditable approach to explainable AI than generative large language models. By proactively quantifying feature importance and uncertainty through the EDAQ framework, the system achieves both high predictive accuracy (89.4%) and substantially improved clinical trust among end users [7], [19]. The four-layer architecture meets regulatory requirements for auditability and reproducibility, advancing the vision of AI systems that engage in collaborative, verifiable reasoning with human stakeholders [1]–[3].

Future work will extend MedTalk to additional regulated domains including insurance and judiciary, evaluate it on larger and more diverse clinical datasets, and investigate mechanisms for real-time explanation updating as new patient data arrive [5], [20].

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